



FEATURES

- Tension and/or Compression
- Small design easy to mount
- Wide temperature range in option
- Very small size

APPLICATIONS

- Micro mechanic equipment
- Miniature press-fit device
- Robotics regulation
- Small size actuators

XFTC301

Miniature Load Cell

SPECIFICATIONS

- 0-500N to 0-10kN [112.4 lbf to 2.248k lbf]
- High Stiffness
- Threaded Male Mechanical Fitting
- Linearity < 0,5% FS

The **XFTC301** series has been specifically developed to measure tension and compression in static and dynamic applications. The miniature size facilitates testing where space is at premium.

The sensing element is fitted with a fully temperature compensated Wheatstone bridge equipped with high stability micro-machined silicon strain gages. The use of silicon strain gages optimizes the load cell's performance at low ranges and frequencies.

A strain relief spring strengthens the cable output. With two threaded male studs, the **XFTC301** is easily installed in industrial or OEM applications.

On request, Instruction documents can be provided to ease the selection and use of our sensors and provide helpful tips.

PERFORMANCE SPECIFICATIONS (typical values at temperature 23°C)

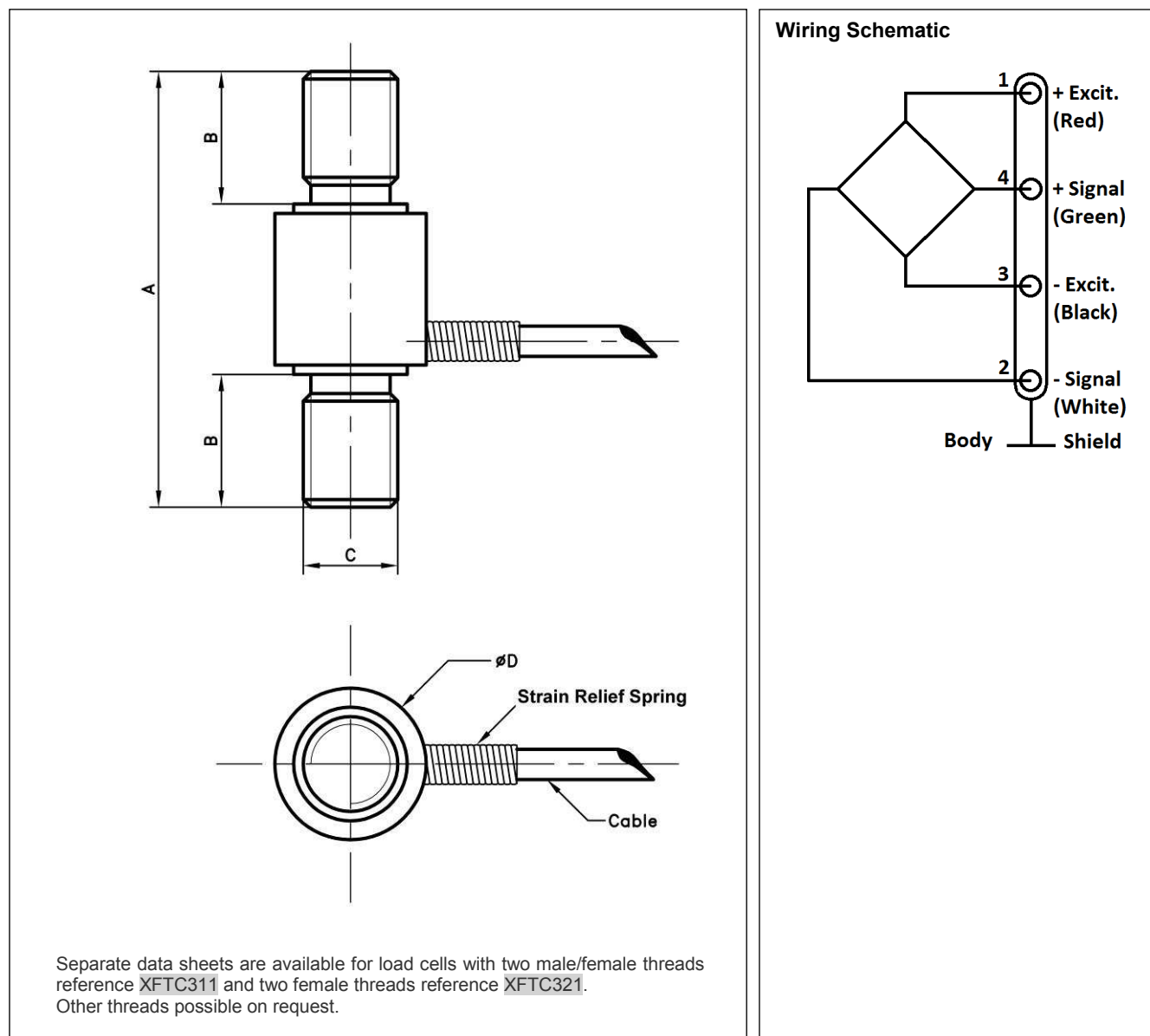
Ranges (FS) (N)	500 N	1 kN	2 kN	5 kN	10 kN
Ranges (lbf)	112.4	224.8	449.6	1124	2248
Material	Aluminium		Stainless steel		
Stiffness (N/m)	1.5E+08	1.5E+08	3.7E+08	4.6E+08	7.8E+08
Stiffness (lbf/ft)	1.0E+07	1.0E+07	2.5E+07	3.2E+07	5.3E+07
Sensitivity (FSO)	70 mV	150 mV	100 mV	150 mV	

Specifications	
Power supply	10Vdc
Overrange Without Damage	2x FS
Overrange Without Destruction	3x FS
Offset	<±10 mV
Input Impedance	1500 ohms
Output Impedance	500 ohms
Linearity	< ±0.5% FS
Hysteresis	< ±0.5% FS
Operating Temperature Range (OTR)	-40°C to +120°C (-40 to 250°F)
Compensated Temperature Range (CTR)	0°C to +60°C (32 to 140°F)
Thermal Zero Shift in CTR	< 2% FS/50°C
Thermal Sensitivity Shift in CTR	< 2%/50°C
Insulation	> 100 Mohms
Protection Index	IP50

Notes

1. Signal goes positive in tension with standard wiring configuration
2. Shielded cable with 4 wires (AWG36/28), standard length 2m [6.5 ft] with strain relief spring
3. Material: Body in stainless steel or aluminum alloy depending on F.S., ; Two male threads M5 or [10-32 UNF], M10 or [3/8-24 UNF] depending on F.S. (metric thread is standard)
4. CE conformance according to EN 61010-1, EN 50081-1, EN 50082-1

DIMENSIONS & WIRING SCHEMATIC (IN METRIC AND IMPERIAL)

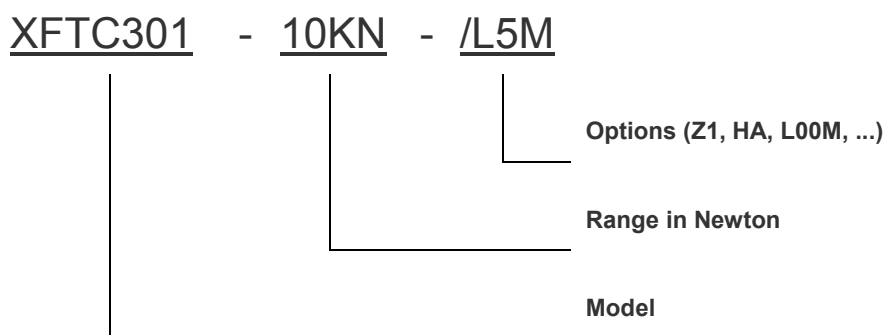


Dimensions in mm [inch]

Full Scale Range in N	500 – 1000	2000	5000 – 10000
A	36 [1.42]		46 [1.81]
B	12.5 [0.49]		14 [0.55]
C (Thread)	M5		M10
Ø D	10 [0.39]		16 [0.63]

OPTIONS

Z0	CTR -40°C to +20°C (-40°F to 68°F)
Z04	CTR -40°C to +90°C (-40°F to 194°F)
Z1	CTR -20°C to +40°C (-4°F to 104°F)
Z3	CTR +20°C to +80°C (68°F to 176°F)
Z35	CTR +20°C to +120°C (68°F to 248°F)
Z36	CTR +20°C to +150°C (68°F to 302°F) - Only available for range 2kN / 5kN / 10kN
HA	Accuracy (CNL&H) $\leq \pm 0.5\%$ F.S.
L5M, L10M, L15M	Special cable length (ex : L5M = 5m length) (Standard cable length 2m)

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